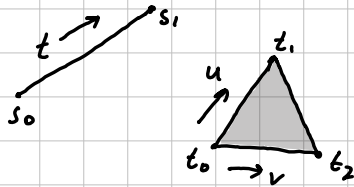


segment-triangle intersection

segment

$$p = s_0 + t(s_1 - s_0) \quad \text{segment} \quad 0 < t < 1$$



triangle

$$p = t_0 + u(t_1 - t_0) + v(t_2 - t_0)$$

$$\text{unit triangle} \quad 0 < u < 1, 0 < v < 1, 0 < u + v < 1$$

set equal

$$s_0 + t(s_1 - s_0) = t_0 + u(t_1 - t_0) + v(t_2 - t_0)$$

$$\underbrace{t(s_1 - s_0)}_a + \underbrace{u(t_0 - t_1)}_b + \underbrace{v(t_2 - t_0)}_c = \underbrace{t_0 - s_0}_d$$

matrix form

$$ta_x + ub_x + vc_x = d_x$$

$$ta_y + ub_y + vc_y = d_y$$

$$ta_z + ub_z + vc_z = d_z$$

$$A = \begin{bmatrix} a_x & b_x & c_x \\ a_y & b_y & c_y \\ a_z & b_z & c_z \end{bmatrix} \quad x = \begin{bmatrix} t \\ u \\ v \end{bmatrix} \quad b = \begin{bmatrix} d_x \\ d_y \\ d_z \end{bmatrix} \quad Ax = b$$

$$\det A = a_x(b_y c_z - b_z c_y) + a_y(b_z c_x - b_x c_z) + a_z(b_x c_y - b_y c_x)$$

$$= a \cdot b \times c \rightarrow \text{triple product!}$$

no sol if $\det A = 0$ (parallel)

cramer's rule

$$x_i = \frac{\det A_i}{\det A}$$

$$A_1 = [d \ b \ c]$$

$$A_2 = [a \ d \ c]$$

$$A_3 = [a \ b \ d]$$

$$t = \frac{d \cdot b \times c}{\det A}$$

$$u = \frac{a \cdot d \times c}{\det A}$$

$$v = \frac{a \cdot b \times d}{\det A}$$

